Study of Radioactive Source Monitoring Based on ElasticNet Regression Method

Bachelor's Thesis Defense Report

Wang Kaiqi

Sino-French Institute of Nuclear Engineering and Technology, Sun Yat-sen University

April 2026

Sun Yat-sen University

Supervisors: Prof. Shi Wei (CityU HK) & Prof. Wang Kai (SYSU)

1. Research Background & Motivation
2. Literature Review
3. Theoretical Basis
4. Algorithm Implementation & Optimization
5. Algorithm Implementation & Optimization
6. PHITS Monte Carlo Simulation
7. Results & Analysis

SYSU

1 Research Background & Motivation

2 Literature Review

3 Theoretical Basis

4 Algorithm Implementation & Optimization

5 Algorithm Implementation & Optimization

6 PHITS Monte Carlo Simulation

7 Results & Analysis

SYSU

Research Background

Nuclear Facility Decommissioning

- Global decommissioning projects increasing (Fukushima Daiichi, etc.)
- IAEA: Accurate source distribution is prerequisite for optimized decommissioning
- Ensuring personnel safety and minimizing waste generation

Three-Dimensional Complexity in Confined Spaces

- **Composite source types**: localized hot-spots + continuous activation zones + dispersed media
- **Nuclide diversity**: ^{134}Cs, ^{137}Cs, ^{60}Co, ^{131}I with distinct energy spectra
- **Measurement constraints**: limited detectors, limited time, dose exposure limits

Problem Statement

The Inverse Problem

- **Forward**: Given sources **s**, predict detector readings **d**
- **Inverse**: Given readings **d**, reconstruct sources **s**

Definition 1.1 Mathematical Formulation

$$\mathbf{d} = A \cdot \mathbf{s} + \varepsilon$$

where **d** = measurements, A = contribution matrix, **s** = source vector, ε = noise.

Key Challenges

- **Underdetermined**: fewer detectors than unknowns ($m \ll n$)
- **Ill-posed**: noisy data leads to unstable solutions

1 Research Background & Motivation

2 Literature Review

3 Theoretical Basis

4 Algorithm Implementation & Optimization

5 Algorithm Implementation & Optimization

6 PHITS Monte Carlo Simulation

7 Results & Analysis

SYSU

Existing Methods and Limitations

Method	Sparse	Continuous	Multi-nuclide	Speed	Core Limitation
ML-EM/MPMI	Fair	Good	Poor	Slow	$O(n^3)$; noise sensitive
IRT-DEGL	Fair	Fair	Fair	V.Slow	Global optimization cost
LASSO	**Excellent**	Poor	Good	Fast	Over-compresses continuous
Fused LASSO	Fair	Good	Good	Medium	Needs neighborhood graph
ElasticNet	**Excellent**	**Excellent**	Good	Fast	Not yet applied to nuclear

Table 1: Comparison of radioactive source reconstruction methods

Why ElasticNet?

Advantages over LASSO and Fused LASSO

- LASSO: discrete/sparse distribution → precision loss for continuous regions
- Fused LASSO: requires predefined neighborhood graph + ADMM convergence issues
- ElasticNet: L1+L2 synergy without extra spatial constraints

Cross-Domain Evidence

- Nuclear medicine (SPECT): outperforms pure L1, L2, fixed-weight EN
- XLCT imaging: superior localization + dual-source resolution
- Fluorescence molecular tomography: 30%+ error reduction

Theorem 2.1 Research Gap

ElasticNet has **not been applied** to nuclear radioactive source monitoring — this thesis fills that gap.

1 Research Background & Motivation

2 Literature Review

3 Theoretical Basis

4 Algorithm Implementation & Optimization

5 Algorithm Implementation & Optimization

6 PHITS Monte Carlo Simulation

7 Results & Analysis

SYSU

ElasticNet Regression Formulation

Definition 3.1 Objective Function (Zou & Hastie, 2005)

$$\min_{\mathbf{w}} \quad \|\mathbf{y} - X\mathbf{w}\|_2^2 + \alpha \left[\rho \|\mathbf{w}\|_1 + \frac{1-\rho}{2} \|\mathbf{w}\|_2^2 \right]$$

L1 vs. L2 Penalty Roles

- **L1** ($\rho\|\mathbf{w}\|_1$): promotes sparsity, drives irrelevant coefficients to zero $\rightarrow$ hot-spot localization
- **L2** ($\frac{1-\rho}{2}\|\mathbf{w}\|_2^2$): groups correlated variables, suppresses oscillation $\rightarrow$ continuous region reconstruction
- $\rho \rightarrow 1$: degrades to LASSO; $\rho \rightarrow 0$: degrades to Ridge

Key Properties

Physical Interpretation

L1 Penalty Physical Meaning

- Count-type penalty on “how many grid cells contain sources”
- Solution tends toward few non-zero concentrated elements — sparse hot-spot localization

L2 Penalty Physical Meaning

- Suppresses individual coefficient amplitude, promotes neighboring cells to have similar values
- Inside continuous activation regions: naturally smooth reconstruction (not “discrete spots” like LASSO)

Theorem 3.1 Synergy Effect

1 Research Background & Motivation

2 Literature Review

3 Theoretical Basis

4 Algorithm Implementation & Optimization

5 Algorithm Implementation & Optimization

6 PHITS Monte Carlo Simulation

7 Results & Analysis

SYSU

Algorithm Pipeline

Stage 1: Preprocessing

- Column normalization: scale each column of A to unit L_2 norm
- Non-negative SVD matrix construction
- Condition number assessment for adaptive α range

Stage 2: ElasticNet Optimization

- `ElasticNet(positive=True)` — hard non-negativity constraint
- GridSearchCV: $\alpha \in [10^{-7}, 10^{0}]$ (50 values), $\rho \in [0.01, 0.99]$ (80 values)
- 5-fold cross-validation with negative MSE scoring
- Custom intercept penalty for $|\beta_0/\bar{y}| > 10\%$

Stage 3: Post-processing

Key Optimization Strategies

Adaptive α Search Range

- cond $= 100$: $\alpha \in [10^{-6}, 10^{0}]$
- cond $= 500$: $\alpha \in [10^{-5}, 10^{1}]$
- cond $= 1000$: $\alpha \in [10^{-4}, 10^{2}]$
- Optimal `l1_ratio` $\approx 0.99 \rightarrow$ strong sparsity prior confirmed

Non-negative Constraint Enhancement

- `positive=True`: hard non-negative constraint
- Non-negative SVD matrix construction eliminates negative response coefficients
- "Non-negativity + Sparse Selection + Smooth Anti-noise" triple mechanism

Intercept Fitting Analysis

1 Research Background & Motivation

2 Literature Review

3 Theoretical Basis

4 Algorithm Implementation & Optimization

5 Algorithm Implementation & Optimization

6 PHITS Monte Carlo Simulation

7 Results & Analysis

SYSU

Algorithm Pipeline

Stage 1: Preprocessing

- Column normalization: scale each column of A to unit L_2 norm
- Non-negative SVD matrix construction
- Condition number assessment for adaptive α range

Stage 2: ElasticNet Optimization

- `ElasticNet(positive=True)` — hard non-negativity constraint
- GridSearchCV: $\alpha \in [10^{-7}, 10^{0}]$ (50 values), $\rho \in [0.01, 0.99]$ (80 values)
- 5-fold cross-validation with negative MSE scoring
- Custom intercept penalty for $|\beta_0/\bar{y}| > 10\%$

Stage 3: Post-processing

Key Optimization: Column Normalization

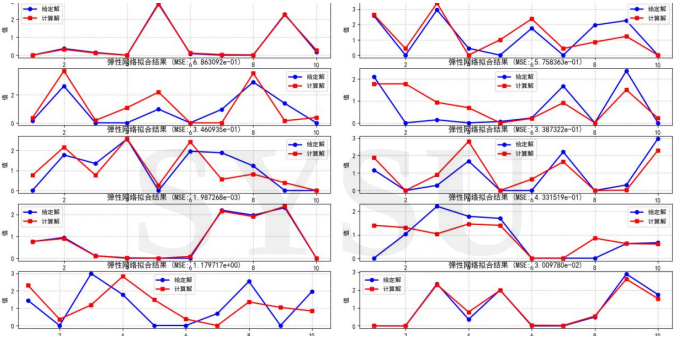


Figure 1: Column normalization improves ElasticNet fitting: cond=100, 5 independent runs all achieve MSE=10^{-4} to 10^{-9}

Non-negative SVD + Non-negative ElasticNet

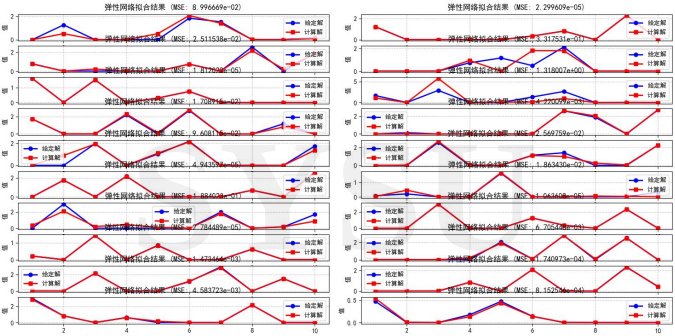


Figure 2: Combined non-negative SVD matrix + non-negative ElasticNet optimization: Exp1/Exp3 reach MSE=10^{-9}

Intercept Fitting Analysis

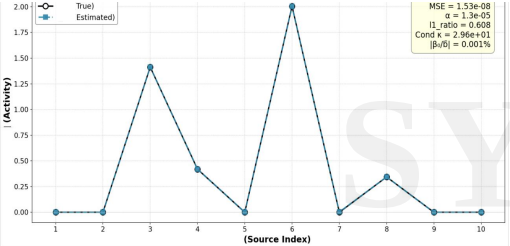


Figure 3: Good case: small intercept, accurate reconstruction

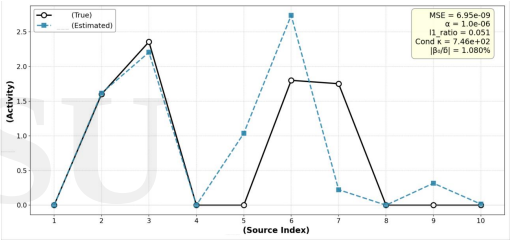


Figure 4: Bad case: large intercept, degraded reconstruction

1 Research Background & Motivation

2 Literature Review

3 Theoretical Basis

4 Algorithm Implementation & Optimization

5 Algorithm Implementation & Optimization

6 PHITS Monte Carlo Simulation

7 Results & Analysis

SYSU

PHITS Simulation Configuration

PHITS (JAEA): Monte Carlo Particle Transport

- 10^8 particles $\times$ 100 batches for statistical precision
- Cs-137 point sources, 0.662 MeV γ-ray

Parameter	Value
Simulation space	Spherical (R=10m, reactor-centered)
Radioactive sources	10 Cs-137 point sources, 1 Ci each
Detectors	8 spherical detectors (R=25mm) at $z = 20$mm
Grid division	$4 \times 3 = 12$ cells
Medium	Air (N, O, Ar at standard density)
Contribution matrix	8×12 (only 11/96 non-zero)
Matrix rank	6/8

Table 2: PHITS simulation parameters

1 Research Background & Motivation

2 Literature Review

3 Theoretical Basis

4 Algorithm Implementation & Optimization

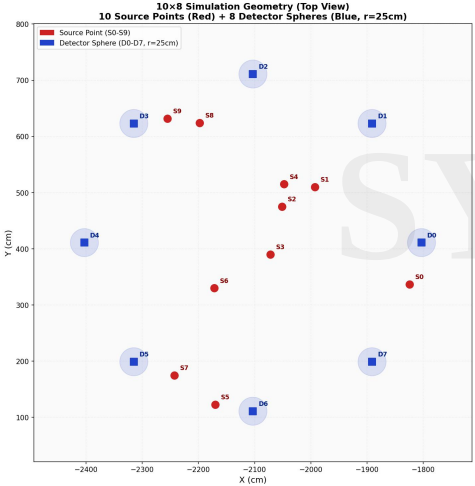
5 Algorithm Implementation & Optimization

6 PHITS Monte Carlo Simulation

7 Results & Analysis

SYSU

Source-Detector Geometry



Configuration

- 10 Cs-137 point sources (1 Ci)
- $4 \times 3 = 12$ grid cells
- 8 spherical detectors (R=25mm)
- 7 active cells, 5 empty
- Matrix: 8×12, rank=6/8

PHITS Validation Results

Best Parameters: $\alpha = 1.0 \times 10^{-6}$, `l1_ratio` $= 0.99$, $R^2 = 0.463$

Cell	True Activity (GBq)	Estimate (GBq)	Rel. Error
Cell 1	1.265×10^5	1.265×10^5	**0.0%**
Cell 5	5.266×10^4	3.93×10^2	99.3%
Cell 6	1.192×10^5	2.31×10^3	98.1%
Cell 7	5.894×10^4	5.894×10^4	**0.0%**
Cell 8	5.724×10^4	5.724×10^4	**0.0%**
Cell 9	4.870×10^4	4.870×10^4	**0.0%**
Cell 10	1.352×10^5	1.373×10^5	**1.2%**

Table 3: 7 non-zero cells detected, 5 with $< 2\%$ error

Reconstruction Visualization

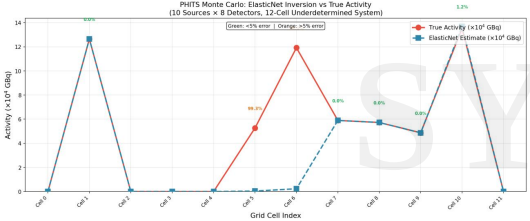


Figure 6: True vs. estimated source distribution

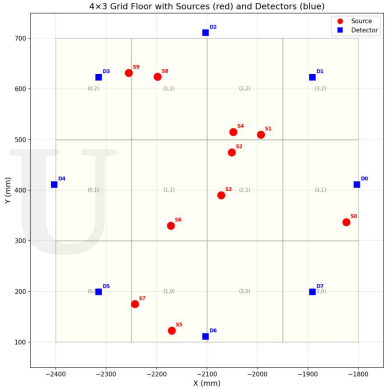


Figure 7: Contribution matrix heatmap (8×12)

Error Source Analysis

Cell 5 & Cell 6 Failure — Physical, Not Algorithmic

- Cell 6: only Detector 1 responds, extremely weak signal (10^{-9} level)
- Cell 5: only Detector 5 responds, weak signal (10^{-10} level)
- Both overwhelmed by stronger responses from other cells

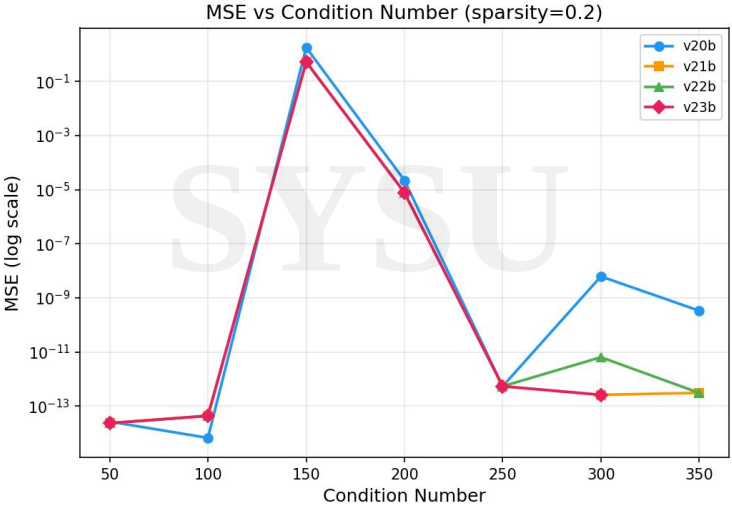
SYSU

Theorem 7.1 Matrix Rank Analysis

- Numerical rank of A = 6/8
- Only 6 of 12 cells algebraically distinguishable
- Physical indistinguishability from detector geometry, **not algorithm failure**

Implication

Condition Number Impact



Bayesian Optimization Results

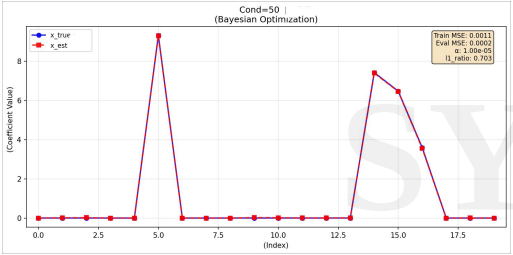


Figure 9: Bayesian optimization: sparse source reconstruction

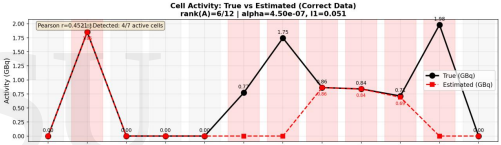


Figure 10: Bayesian + PHITS Monte Carlo validation

Condition Number & Bayesian Optimization

Difficulty Inflection Points at Cond ≈ 150 and 300

- Not simply "more ill-conditioned = harder" — critical spectral structure
- All optimization strategies show diminishing returns at high condition numbers
- Optimal `l1_ratio` ≈ 0.99 universally — confirms strong sparsity prior

	Grid Search	**Bayesian Opt.**
Evaluations	150–450	40
Computation time	Baseline	5–10× faster
Accuracy (low cond)	Reference	Comparable
Accuracy (high cond)	**Higher ceiling**	Slightly worse

Table 4: Grid search vs. Bayesian optimization — complementary, not substitutive

1 Research Background & Motivation

2 Literature Review

3 Theoretical Basis

4 Algorithm Implementation & Optimization

5 Algorithm Implementation & Optimization

6 PHITS Monte Carlo Simulation

7 Results & Analysis

SYSU

Conclusions

1. ElasticNet fills the application gap

Effectively handles “sparse + continuous” composite source reconstruction in nuclear monitoring.

2. Optimization strategies improve accuracy significantly

Column normalization + adaptive α + non-negative constraint + intercept penalty.

3. PHITS validation confirms practical value

All 7 non-zero cells detected; 5 cells with $< 2\%$ error.

4. Bayesian optimization: complementary to grid search

70–90% computation savings; grid search has higher accuracy ceiling.

Future Work

Mobile Detection Platforms (UGV / UAV)

- Carry detectors for large-scale autonomous patrol
- Richer contribution matrix → improved reconstruction stability

Algorithm Improvements

- Adaptive ElasticNet for automatic variable selection
- Bayesian ElasticNet with uncertainty quantification
- Deep learning + ElasticNet model cascade

Extended Applications

- Multi-energy source discrimination
- 3D volumetric source reconstruction

1 Research Background & Motivation

2 Literature Review

3 Theoretical Basis

4 Algorithm Implementation & Optimization

5 Algorithm Implementation & Optimization

6 PHITS Monte Carlo Simulation

7 Results & Analysis

SYSU

References I

[1] Tibshirani R. Regression shrinkage and selection via the lasso. *JRSS-B*, 1996.

[2] Hoerl & Kennard. Ridge regression. *Technometrics*, 1970.

[3] Sugaya et al. Inverse estimation for fuel debris search. *ANE*, 2019.

[4] Shi W et al. LASSO with spectral information. *ANE*, 2023.

[5] Shi W et al. LASSO: Theory and demonstration. *PNE*, 2023.

[6] Yamada et al. Fused LASSO for radioactive materials. *WM2024*, 2024.

[7] Zou & Hastie. Regularization via the elastic net. *JRSS-B*, 2005.

[8] Kasai & Otsuka. Dynamic elastic net for nuclear medicine. *J. Imaging*, 2025.

SYSU

References II

[9] Zhao et al. Robust elastic net for XLCT. *Phys. Med. Biol.*, 2021.

[10] Sato et al. PHITS version 3.02. *JNST*, 2018.

SYSU

1 Research Background & Motivation

2 Literature Review

3 Theoretical Basis

4 Algorithm Implementation & Optimization

5 Algorithm Implementation & Optimization

6 PHITS Monte Carlo Simulation

7 Results & Analysis

SYSU

Questions & Discussion

Thank you for your attention!

Wang Kaiqi | 22355080

Supervisors: Prof. Shi Wei (CityU HK) & Prof. Wang Kai (SYSU)